PHASE-1 REPORT

Data Analytics Track

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Track	Data Analytics
GitHub Repository	https://github.com/vaishnaviraut629-de/pulseiq
Git Tag	phase-1

1. Task Completion Summary

Task	Status	Outcome
Task 1: ETL Pipeline	Completed	Loaded 112 rows into raw_employees.
Task 2: Star Schema Warehouse	Completed	Created dim_employee, dim_department, and fact_headcount.
Task 3: Data Quality Gatekeeper	Completed	All data quality checks passed (dq_checks).
Task 4: Governed Metric Layer	Completed	M01_Headcount metric returned a headcount of 112.
Task 5: Workforce Dashboard	Completed	Created the CEO view in 1_Workforce.py.

2. Validation and Proof of Functionality

The reported headcount was validated against the raw employee data. The query SELECT COUNT(*) FROM M01_Headcount returned 112, matching the 112 rows in raw_employees.csv.

The data quality script, dq_checks.py, reported: “ALL CHECKS PASSED.”

The dashboard application reads only from the M01 metric view and does not query the raw table directly.

3. Mini Challenge: M06_OfficeRisk

Definition: Office Risk / On-site Presence Rate = (Onsite + Hybrid) / Total Employees

Measure	Value
Onsite employees	68
Hybrid employees	23
Total employees	112
Onsite + Hybrid	68 + 23 = 91
Calculation	91 / 112 × 100

Result	81.25%
Time taken	35 minutes

The calculation was manually checked using a calculator: 91 ÷ 112 = 81.25% (rounded to two decimal places).

4. Self-Assessment

1. Trace the headcount of 112 back to the raw data: Yes.

2. Explain the Star Schema: Yes.

5. Key Learnings

- Governed metric views should be treated as the single source of truth for dashboard metrics.
- Dashboard design should use a limited, consistent palette of three colors.

6. Problems Encountered and Resolutions

Issue	Resolution
Path containing spaces	Resolved the path issue by enclosing the path in quotation marks.
Raw table access in dashboard	Updated the dashboard to read from the M01 metric view.
Verification	Sudo verification was completed via email, as reported.

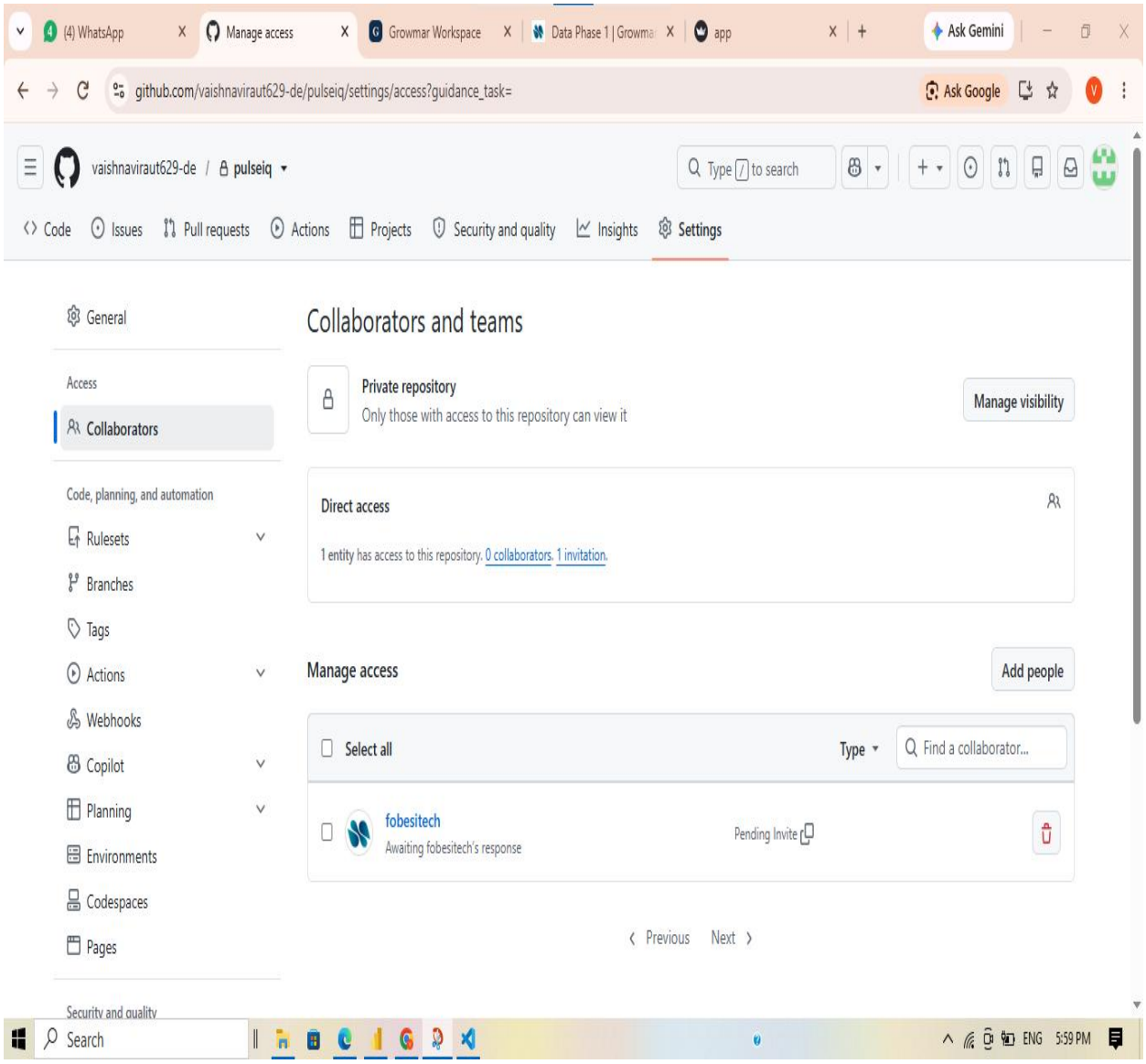
7. Next Phase Plan

Phase 2 will focus on building metrics M02–M05 and adding new dashboard pages.

8. Screenshots and Supporting Evidence

The following evidence items are to be inserted into the final report:

- 08_dashboard.png
- 09_trend.png
- 10_breakdown.png
- GitHub invitation proof
- Git tag proof (phase-1)
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End of Phase-1 Report